

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet

range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1.IP Block: 192.168.20.0/24

Lab A (/26)

Step 1: Subnet mask = /26 = 255.255.255.192

Step 2: Block size = 64

Subnets:

- 192.168.20.0
- 192.168.20.64
- 192.168.20.128
- 192.168.20.192

Assign the first subnet to Lab A.

Network Address: 192.168.20.0

Valid Host Range: 192.168.20.1 to 192.168.20.62

Broadcast Address: 192.168.20.63

Usable Hosts: $2^6 - 2 = 62$ hosts

Lab B (/27)

Step 1: Subnet mask = /27 = 255.255.255.224

Step 2: Block size = 32

Use the next available subnet.

Network Address: 192.168.20.64

Valid Host Range: 192.168.20.65 to 192.168.20.94

Broadcast Address: 192.168.20.95

Usable Hosts: $2^5 - 2 = 30$ hosts

Lab C (/28)

Step 1: Subnet mask = /28 = 255.255.255.240

Step 2: Block size = 16

Use the next available subnet.

Network Address: 192.168.20.96

Valid Host Range: 192.168.20.97 to 192.168.20.110

Broadcast Address: 192.168.20.111

Usable Hosts: $2^4 - 2 = 14$ hosts

2. System 1 and System 2

Subnet Mask = 255.255.255.192 (/26)

Block size = 64

Subnets:

- 192.168.1.0 – 63
- 192.168.1.64 – 127
- 192.168.1.128 – 191
- 192.168.1.192 – 255

System 1

IP Address = 192.168.1.130

Step 1: IP 130 falls between 128 and 191.

Network Address: 192.168.1.128

Valid Host Range: 192.168.1.129 to 192.168.1.190

Broadcast Address: 192.168.1.191

Result: 192.168.1.130 is a valid host in this subnet.

System 2

IP Address = 192.168.1.190

Step 1: IP 190 also falls between 128 and 191.

Network Address: 192.168.1.128

Valid Host Range: 192.168.1.129 to 192.168.1.190

Broadcast Address: 192.168.1.191

Result: 192.168.1.190 is also a valid host in this subnet.

3. IP Block: 172.16.0.0/16

HR (/25)

Subnet Mask: 255.255.255.128

Network Address: 172.16.0.0

Host Range: 172.16.0.1 to 172.16.0.126

Broadcast Address: 172.16.0.127

Usable Hosts: 126

Sales (/26)

Subnet Mask: 255.255.255.192

Network Address: 172.16.0.128

Host Range: 172.16.0.129 to 172.16.0.190

Broadcast Address: 172.16.0.191

Usable Hosts: 62

IT (/27)

Subnet Mask: 255.255.255.224

Network Address: 172.16.0.192

Host Range: 172.16.0.193 to 172.16.0.222

Broadcast Address: 172.16.0.223

Usable Hosts: 30

Admin (/28)

Subnet Mask: 255.255.255.240

Network Address: 172.16.0.224

Host Range: 172.16.0.225 to 172.16.0.238

Broadcast Address: 172.16.0.239

Usable Hosts: 14

Remaining Unused IP Range

Unused addresses start from 172.16.0.240 up to 172.16.255.255

4. IP Block: 172.16.10.0/24 using /27

Step 1: Original network = /24

Subnet mask = /27

Borrowed bits = $27 - 24 = 3$

Total Subnets = $2^3 = 8$

Usable Hosts per Subnet = $2^5 - 2 = 30$

Find the subnet for IP 172.16.10.78

Block size = 32

Subnets:

- 0–31
- 32–63
- 64–95
- 96–127
- 128–159
- 160–191
- 192–223

- 224–255

IP 78 belongs to 64–95.

Network Address: 172.16.10.64

Host Range: 172.16.10.65 to 172.16.10.94

Broadcast Address: 172.16.10.95

Check IP 172.16.10.95

172.16.10.95 is the broadcast address of the subnet.

Result: It belongs to the same subnet but cannot be assigned to a host.

5. IPv6 Address

Given Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the Address

Remove leading zeros and replace consecutive zero blocks with "::".

Compressed Address:

2001:db8::ff00:42:8329

Step 2: Expand the Address

Expanded Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Find the Network Prefix (/64)

A /64 prefix means the first 64 bits (first four groups) are the network.

Network Prefix:

2001:db8:0:0::/64

Step 4: Total Host Addresses

Host bits = $128 - 64 = 64$

Total possible host addresses = 2^{64}

Answer: 18,446,744,073,709,551,616 possible host addresses.